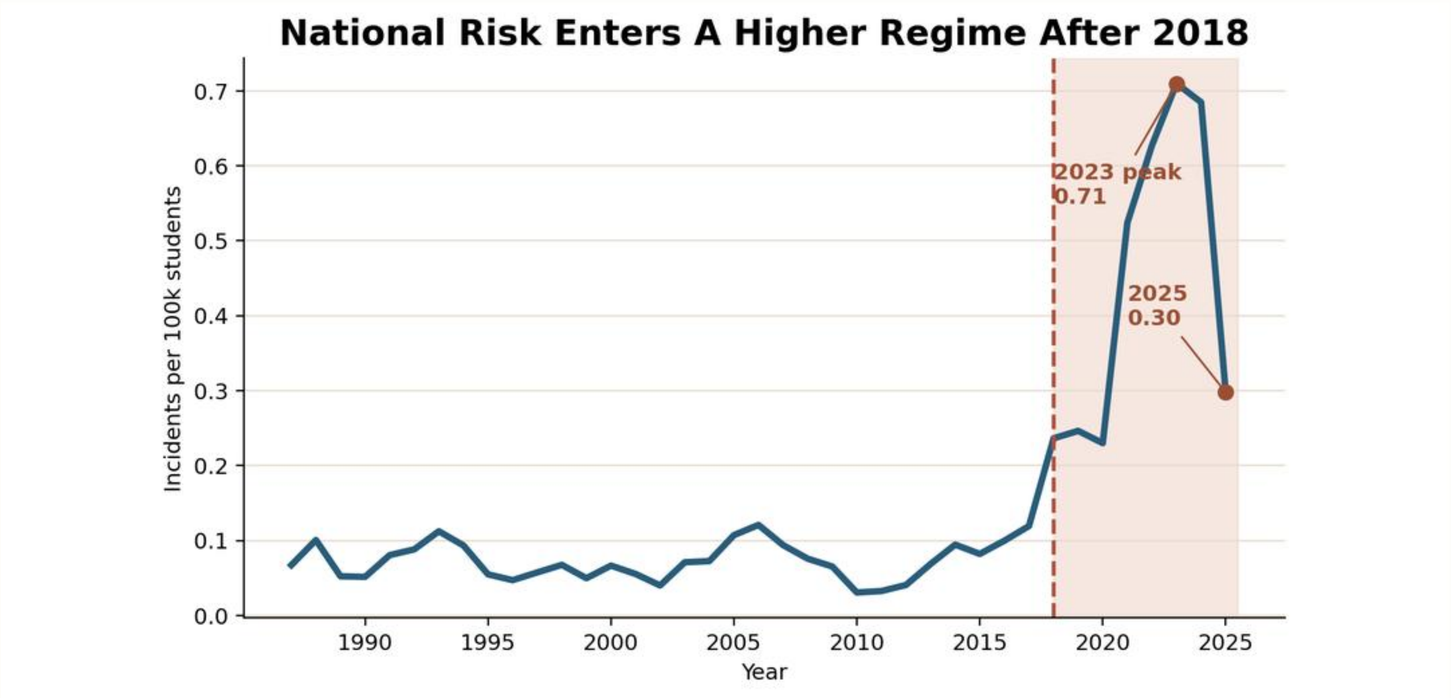
Title

Rising National Risk Is The Main Finding

This deck uses only the project’s saved data, model outputs, and existing analytic conclusions. The strongest supported result is a sharp post-2018 increase in national incident risk.



0.71

Peak national rate in the project tables, reached in 2023.

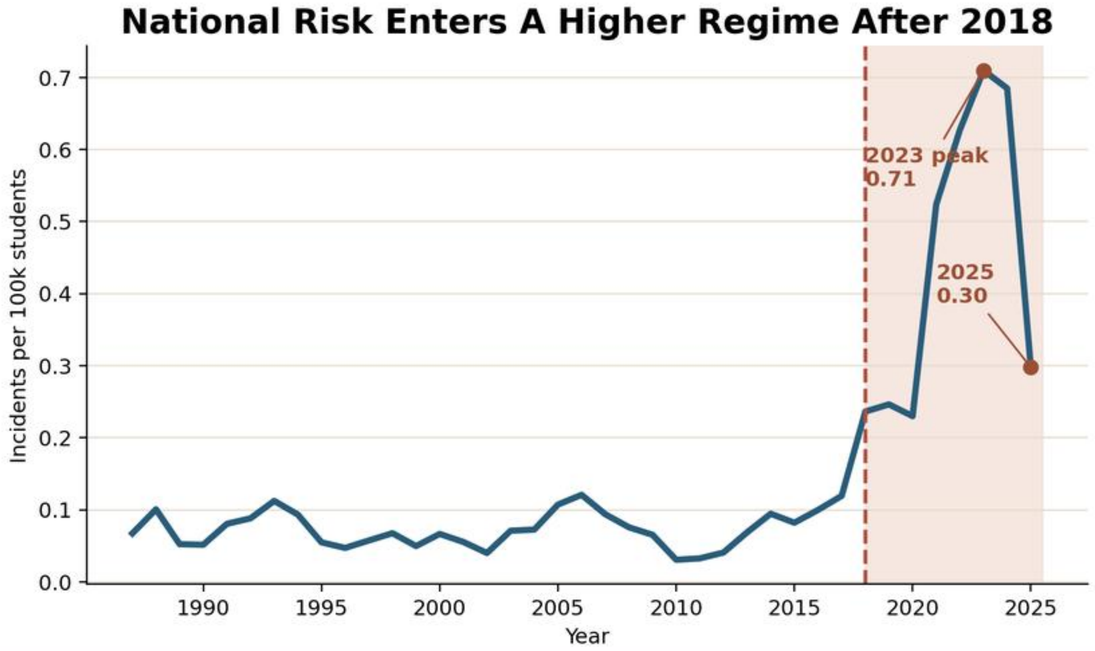
Main finding

National incident risk rises sharply after 2018 and remains far above most earlier years in the available 1987-2025 series.

Core Insight

National Risk Enters A Higher Post-2018 Regime

The clearest project-wide result is not a single coefficient. It is the visible shift in national risk after 2018, peaking in 2023 and staying elevated in later years.

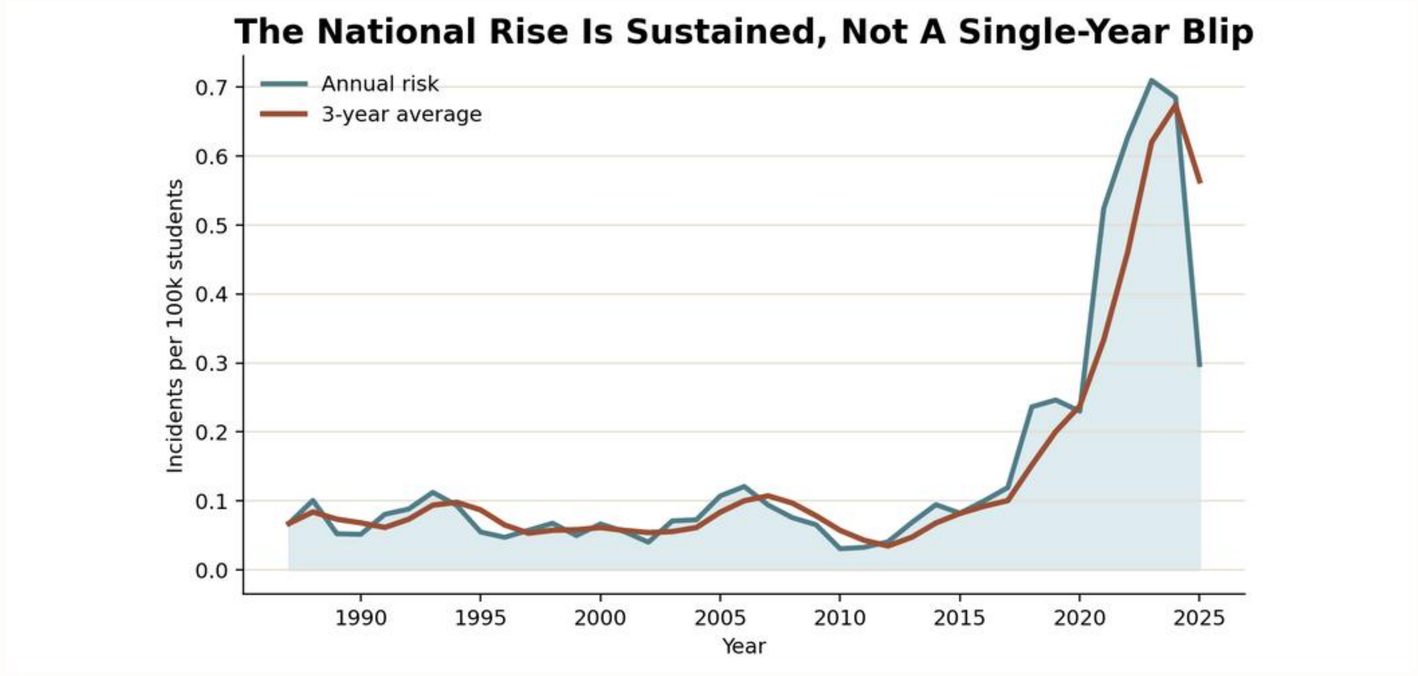


The project tables report 0.709825 in 2023 and 0.297758 in 2025.

National Pattern

The Increase Is Sustained, Not Just A One-Year Spike

The project normalizes incidents by enrollment and explicitly repairs denominator comparability issues. That makes the late-year rise a stronger signal than raw incident counts alone.



1987-2025

Core analytic window used for the cleaned national series.

-0.35%

Latest 2025 enrollment YoY change in the Plotly insight summary.

39

Annual observations in the national series.

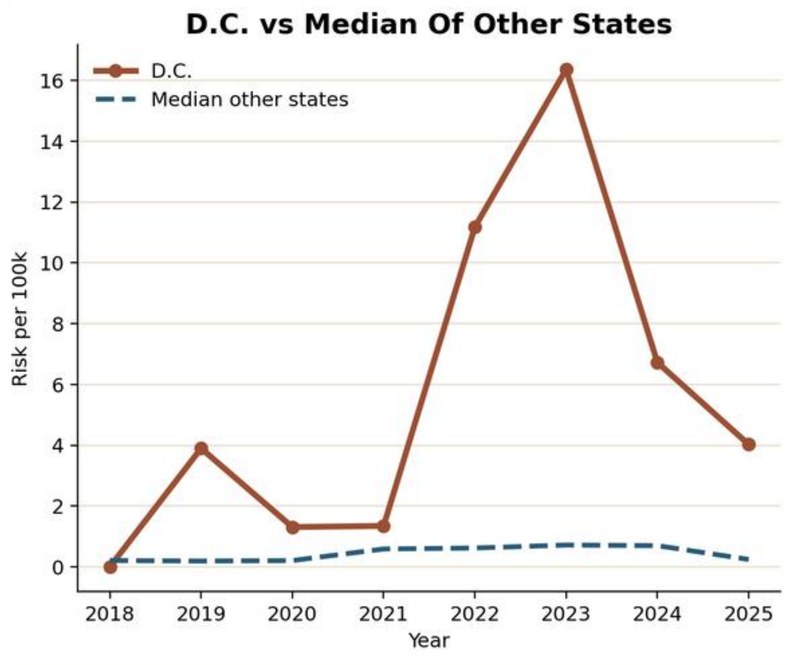
What this proves

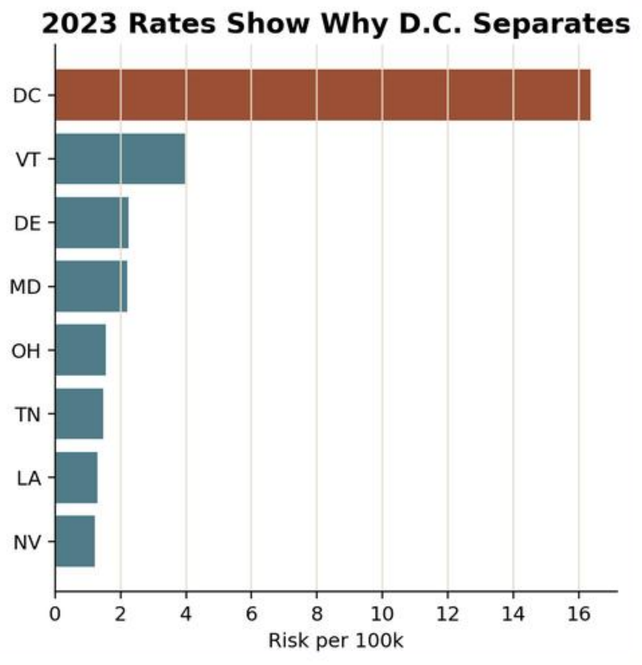
- The national rise survives population normalization because the rate is built on total students, not raw counts.
- The project explicitly flags and repairs the 2019 enrollment comparability break before interpreting trend movement.
- The strongest story in the repo is the broader time pattern, not a broad set of stable policy effects.

D.C. vs Other States

D.C. Looks Extreme Because Short Bursts Hit A Much Smaller Student Base

D.C. is a genuine rate outlier in the project data, but the reason is visible once incident counts and student exposure are considered together.





12
D.C. incidents in 2023.

73,364
D.C. student base in 2023.

16.36
D.C. risk in 2023.

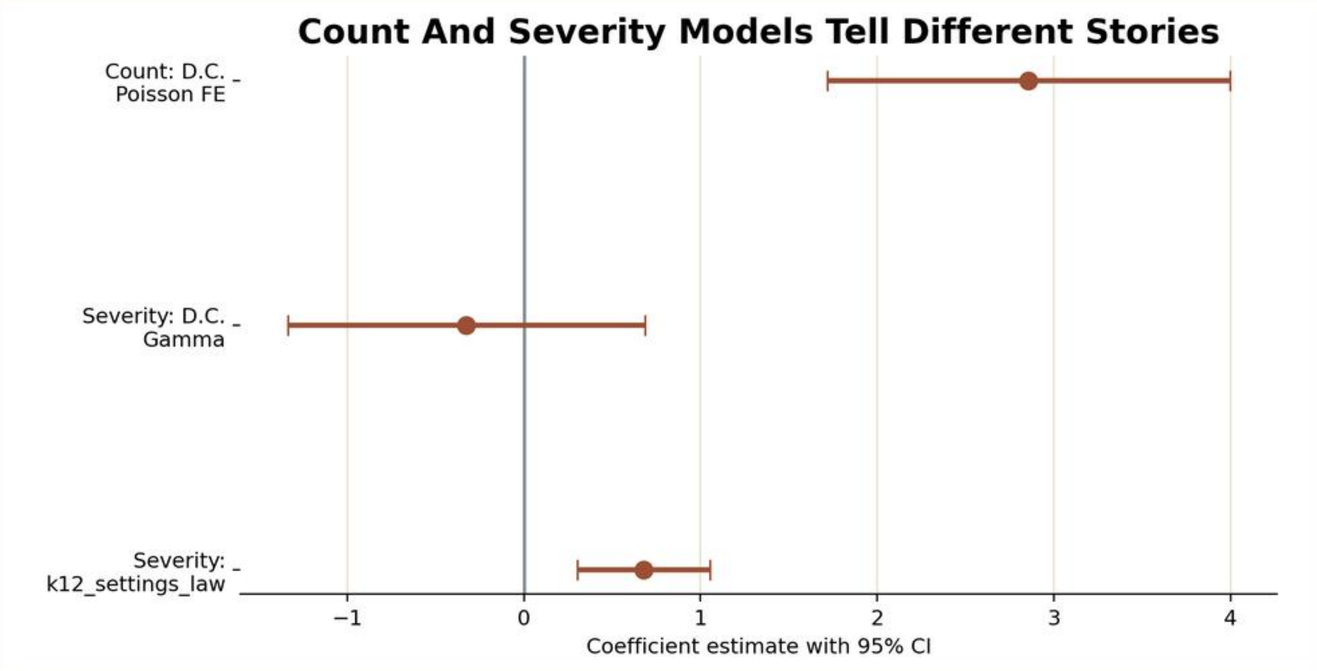
Why D.C. separates

- The notebooks report D.C. at 11.175213 in 2022 and 16.356796 in 2023 before falling to 4.025333 in 2025.
- The 2023 spike is 12 incidents over 73,364 students, so the denominator is small enough to magnify a short burst into a very high rate.
- That makes D.C. a special case in rate space without overturning the broader national trend.

Model Results

The Final Count Model Flags D.C., But The Severity Model Does Not

The project’s final count specification is Poisson with state and year fixed effects, a log enrollment offset, and state-clustered standard errors. The severity model is a Gamma GLM on victims per incident.



Model facts from the repo

- README reports NB2 alpha near zero at about 0.03, statistically indistinguishable from zero, with unstable convergence.
- That is why the final count model is Poisson with clustered state-level standard errors rather than Negative Binomial.
- The count model reports a large positive D.C. effect, while the severity model’s D.C. term crosses zero.

2.8558

Count model: D.C. coefficient with 95% CI [1.716, 3.996].

-0.3264

Severity model: D.C. coefficient with 95% CI [-1.338, 0.686].

0.6781

Severity model: k12_settings_law with 95% CI [0.302, 1.054].

Interpretation: D.C. is a strong rate effect in the count model, but not a conditional-severity outlier once an incident occurs.

Interpretation

The Repo Supports A Trend Story First, A D.C. Outlier Story Second, And A Limited Policy Story Third

The cleaned panel, the final model choice, and the D.C. special-case results line up. The strongest result is rising national risk over time rather than a broad causal policy story.

Data

The project builds a cleaned 1987-2025 panel, preserves zero-incident years, and normalizes incidents by student exposure.

Models

The final count model is Poisson FE with clustered standard errors; the severity model is a Gamma GLM and yields a different D.C. conclusion.

D.C.

D.C. is a rate outlier because short bursts operate over a smaller denominator, but it is not a significant severity outlier in the Gamma model.

Bottom line: the project’s strongest evidence points to a rising national risk regime, with D.C. as an important but non-decisive speci

Conclusion

Rising National Risk Is The Cleanest Final Takeaway

The full project is most persuasive when read as a cleaned, enrollment-normalized national risk story with D.C. treated as a distinct but not decisive outlier.

0.71 in 2023

Peak national risk reported in the project tables.

D.C. rate outlier

Large count-model state effect, but no significant severity effect in the Gamma model.

Policy signals are limited

The broader time pattern is stronger and more stable than most individual policy coefficients.

Final read

Monitor the national trend first.

Closing takeaway

- National incident risk rises sharply after 2018 and reaches its peak in 2023.
- D.C. appears different because a small student base magnifies short bursts into extreme per-capita rates.
- The final count model supports a strong D.C. risk effect, but the severity model does not.
- The strongest evidence in the repo supports monitoring the time trend more than claiming broad uniform policy effects.